

ADIL JASEEM

FULL STACK SOFTWARE ENGINEER | AI PLATFORMS, DISTRIBUTED SYSTEMS & LLM INFRASTRUCTURE

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 3+ years of professional experience building enterprise software across product interfaces, Python microservices, event-driven workflows, real-time communication, data systems, observability, and platform infrastructure. Built QuantiFore's complete React and TypeScript frontend and core services including SML Coordinator, SML Orchestrator, Query Parser, Entity Resolution, WebSocket Gateway, Signal Scraper, and Signal Scheduler. Experienced with gRPC, RabbitMQ, PostgreSQL, TimescaleDB, graph databases, Docker Compose, Grafana/Loki/Alloy, and self-hosted LLM infrastructure using llama.cpp, Ollama, MLX, and OpenAI-compatible endpoints.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Frontend: React.js, Next.js, Vite, Tailwind CSS, React Router, Axios, Framer Motion, Mapbox GL, D3.js, Material UI

Backend & APIs: FastAPI, Node.js, GraphQL, gRPC, Protocol Buffers, REST APIs, WebSockets, Pydantic, SQLAlchemy, asyncpg, asyncio, Alembic

Architecture & Data: Microservices, event-driven systems, RabbitMQ, PostgreSQL, TimescaleDB, Neo4j, Apache AGE, Redis, vector search, Pinecone

AI & LLM Infrastructure: llama.cpp, Ollama, MLX, GGUF models, OpenAI-compatible APIs, Sentence Transformers, OpenAI, Gemini, Claude, MCP, Crawl4AI, Wikidata

Platform & Observability: Docker, Docker Compose, Bash, Git/Gitflow, Grafana, Loki, Grafana Alloy, Authentik, Okta, Kong Gateway, Poetry

PROFESSIONAL EXPERIENCE

Full Stack Developer | QuantiFore | Numenor

February 2025 - Present

Kochi, Kerala, India

- Built QuantiFore's complete React and TypeScript frontend across authentication, onboarding, dashboard analytics, monitoring, signal analysis, alerts, causal visualization, simulation, model management, administration, settings, API integration, and real-time workflows.
- Designed and implemented the core Stock or Model Lookup workflow, including the SML Coordinator and SML Orchestrator services, using gRPC, RabbitMQ, PostgreSQL, Redis, asynchronous processing, semantic cache checks, workflow state management, and downstream service coordination.
- Built the Query Parser and Entity Resolution services for natural-language intent detection, entity extraction, anchor/driver identification, semantic tokens and embeddings, canonical naming, alias resolution, contextual disambiguation, and Wikidata/local-LLM integration.
- Developed the WebSocket Gateway Service to deliver authenticated SML completion events, alerts, job updates, and real-time platform notifications to frontend sessions.
- Architected and developed the Signal Scraper and Signal Scheduler services, covering LLM query planning, trusted-source ranking, relevance screening, Crawl4AI extraction, evidence verification, geocoding, deduplication, job claiming, stale-claim recovery, persistence, and alert publication.
- Set up self-hosted model-serving infrastructure using llama.cpp, Ollama, MLX, GGUF models, and OpenAI-compatible endpoints, including model routing, context/concurrency configuration, retries, fallbacks, and integrations with application and developer workflows.
- Built and maintained development, staging, and production workflows for a 26+ container platform using Docker Compose, environment-specific configuration, health checks, networks, volumes, service dependencies, startup ordering, and bootstrap automation for lifecycle management and troubleshooting.
- Configured centralized observability with Grafana, Loki, and Grafana Alloy, and maintained Git branching, repository organization, service integration, environment coordination, merge workflows, and release-oriented platform operations.

PROFESSIONAL EXPERIENCE (CONTINUED)

Software Developer | Grapeline Innovations Private Limited
Kochi, Kerala, India

December 2022 - September 2024

- Developed reusable React.js interfaces, responsive layouts, and interactive product workflows across Newsraven, Clinasyst, and Responseloop.
- Built Node.js and GraphQL backend functionality, including API operations, scheduled tasks, server-side processing, PostgreSQL integration, authentication flows, and Google Maps-based features.
- Integrated OpenAI, Gemini, and Claude for summarization, question answering, AI-generated content, semantic search, and similarity-search workflows using PostgreSQL, vector databases, and Pinecone.
- Performed code reviews, removed redundant code, supported test deployments, validated features, and collaborated across frontend and backend delivery.

SELECTED ENGINEERING WORK

QuantiFore Platform - Full-Stack & Platform Engineering
Built the complete analytical frontend, core natural-language model lookup services, WebSocket delivery, signal-intelligence services, shared local-model integrations, Docker environments, bootstrap workflows, observability, and Git coordination for an event-driven intelligence and system-dynamics platform.
Technology: React, TypeScript, Python, FastAPI, gRPC, RabbitMQ, PostgreSQL, TimescaleDB, Apache AGE, Redis, WebSockets, Docker, Grafana, llama.cpp, Ollama

Evidence-Backed Signal Intelligence Pipeline
Created an asynchronous discovery pipeline that plans searches, ranks trusted sources, screens relevance in batches, crawls selected pages, extracts structured signals, verifies evidence against source content, geocodes locations, deduplicates results, persists state, and publishes completion and alert events.
Technology: Python, aio-pika, asyncpg, SearXNG, Crawl4AI, MCP, Wikidata, local LLM endpoints, PostgreSQL

Stock or Model Lookup Workflow
Implemented the path from user query to ready model through immediate job acceptance, organization-context hydration, typed parsing, entity resolution, exact and semantic reuse, fast/slow workflow branching, asynchronous coordination, and real-time completion notification.
Technology: Python, gRPC, Protocol Buffers, RabbitMQ, PostgreSQL, Redis, Sentence Transformers, WebSockets

ADDITIONAL PLATFORM EXPERIENCE

- Designed and integrated versioned gRPC and Protocol Buffer service contracts, RabbitMQ direct/topic exchanges, dead-letter handling, publisher confirmations, trace context, and asynchronous job envelopes.
- Worked across PostgreSQL, TimescaleDB, Apache AGE/Neo4j, Redis, async connection pools, Alembic migrations, semantic/vector lookup, and application-oriented multi-database integration.
- Configured and integrated Authentik, Okta, JWT/OIDC flows, Kong gateway policies, CDC/outbox workflows, health checks, structured JSON logging, and production troubleshooting through centralized logs.
- Collaborated with and supported ETL, CFRI, ATI, blueprint, dashboard, simulation, and reporting services while preserving accurate ownership boundaries for specialist implementations.

EDUCATION & EARLY CAREER

Bachelor of Computer Applications - Gems Arts and Science College, Ramapuram
MEARN Stack Web Development - Luminar Technolab, Kochi

2019 - 2022
May - Sep 2022

Additional achievement: Participant in the Guinness World Records achievement for the largest picture mosaic formed by 4,500 students (2017).